

An Exact Divisor-Theoretic Classification of the Integer Solutions of

$$z^2 + y^2z + x^3 - 2 = 0$$

Abstract

We give a complete elementary classification of the triples $(x, y, z) \in \mathbb{Z}^3$ satisfying

$$z^2 + y^2z + x^3 - 2 = 0.$$

The central observation is not merely a construction but an explicit bijection: a solution corresponds uniquely to an ordered pair of integer factors a, b satisfying $ab = x^3 - 2$ and $a + b = y^2$, through $a = -z$ and $b = z + y^2$. Resolving the signs of these two factors yields a canonical description in two exhaustive cases, $x \geq 2$ and $x \leq 1$. We prove both directions of the classification, its uniqueness statement, the treatment of all diagonal and zero-square cases, a compact ordered-divisor formulation, a root-swapping involution, and a terminating divisor algorithm which enumerates all solutions for each fixed value of x .

Keywords. Diophantine equation; integer solutions; divisor parametrization; quadratic factorization; integral points.

1 Introduction and notation

We study the Diophantine equation

$$z^2 + y^2z + x^3 - 2 = 0, \quad (x, y, z) \in \mathbb{Z}^3. \quad (\text{E})$$

The purpose of this note is to describe its entire integer solution set by an exact divisor criterion. The word “exact” is important: every triple produced by the criterion is verified to satisfy (E), and every integer solution of (E) is proved to arise from the criterion with uniquely determined factor data, up to the explicitly stated coincidences.

We use

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}.$$

For $s \in \mathbb{Z}_{\geq 0}$, define

$$\mathcal{Y}(s) = \begin{cases} \{0\}, & s = 0, \\ \{-s, s\}, & s > 0. \end{cases} \quad (1)$$

Thus $\mathcal{Y}(s)$ is exactly the set of integers whose absolute value is s ; the definition avoids counting $+0$ and -0 as different values.

Let

$$\mathcal{S} = \{(x, y, z) \in \mathbb{Z}^3 : z^2 + y^2z + x^3 - 2 = 0\}$$

and, for a fixed $x \in \mathbb{Z}$, let

$$\mathcal{S}_x = \{(x, y, z) \in \mathcal{S}\}.$$

The factorization driving the entire argument is

$$(-z)(z + y^2) = x^3 - 2. \quad (2)$$

The main work is to formulate this identity as a bijection and then resolve all possible signs without losing or duplicating solutions.

2 The exact factor-pair correspondence

We begin by ruling out a degenerate value which could otherwise make one of the factors in (2) vanish.

Lemma 2.1. *For every integer x , one has $x^3 \neq 2$. Consequently, if $(x, y, z) \in \mathcal{S}$, then*

$$z \neq 0 \quad \text{and} \quad z + y^2 \neq 0.$$

Proof. If $x \leq 1$, then $x^3 \leq 1 < 2$. If $x \geq 2$, then $x^3 \geq 8 > 2$. Since every integer lies in exactly one of these two cases, $x^3 = 2$ has no integer solution.

Now suppose $(x, y, z) \in \mathcal{S}$. Rearranging (E) gives

$$z(z + y^2) = 2 - x^3.$$

By the first part, the right-hand side is nonzero. A product of two integers is nonzero only when both factors are nonzero, so $z \neq 0$ and $z + y^2 \neq 0$. $\square$

The next proposition is the precise logical core of the classification.

Proposition 2.2 (Factor-pair bijection). *Let*

$$\mathcal{T} = \{(x, y, a, b) \in \mathbb{Z}^4 : ab = x^3 - 2 \text{ and } a + b = y^2\}.$$

Define

$$\begin{aligned} \Phi : \mathcal{S} &\longrightarrow \mathcal{T}, & \Phi(x, y, z) &= (x, y, -z, z + y^2), \\ \Psi : \mathcal{T} &\longrightarrow \mathcal{S}, & \Psi(x, y, a, b) &= (x, y, -a). \end{aligned}$$

Then Φ and Ψ are well-defined mutually inverse bijections. In particular, for every solution (x, y, z) , the associated ordered factor pair is uniquely

$$a = -z, \quad b = z + y^2. \quad (3)$$

Proof. First let $(x, y, z) \in \mathcal{S}$ and put $a = -z$ and $b = z + y^2$. Then

$$a + b = -z + z + y^2 = y^2.$$

Moreover, using (E),

$$ab = (-z)(z + y^2) = -z^2 - y^2z = x^3 - 2.$$

Thus $(x, y, a, b) \in \mathcal{T}$, so Φ is well-defined.

Conversely, let $(x, y, a, b) \in \mathcal{T}$ and put $z = -a$. Since $a + b = y^2$ and $ab = x^3 - 2$, we obtain

$$z^2 + y^2z + x^3 - 2 = a^2 - a(a + b) + ab$$

$$\begin{aligned}
&= a^2 - a^2 - ab + ab \\
&= 0.
\end{aligned}$$

Hence $(x, y, -a) \in \mathcal{S}$, so Ψ is well-defined.

It remains to verify that the two maps are inverse to one another. For $(x, y, z) \in \mathcal{S}$,

$$(\Psi \circ \Phi)(x, y, z) = \Psi(x, y, -z, z + y^2) = (x, y, z).$$

For $(x, y, a, b) \in \mathcal{T}$, the relation $a + b = y^2$ gives

$$\begin{aligned}
(\Phi \circ \Psi)(x, y, a, b) &= \Phi(x, y, -a) \\
&= (x, y, a, -a + y^2) \\
&= (x, y, a, b).
\end{aligned}$$

Therefore Φ and Ψ are mutually inverse bijections. Formula (3) follows directly from the definition of Φ and also shows uniqueness of the ordered factor pair. $\square$

3 Sign analysis and direct verification

The sign restrictions needed below are elementary, but we state and prove them explicitly so that no case is hidden.

Lemma 3.1 (Positive product). *Let $a, b \in \mathbb{Z}$. If $ab > 0$ and $a + b \geq 0$, then $a > 0$ and $b > 0$.*

Proof. The inequality $ab > 0$ implies that a and b have the same sign and are both nonzero. If they were both negative, then $a + b < 0$, contradicting the hypothesis. Hence both are positive. $\square$

Lemma 3.2 (Negative product). *Let $a, b \in \mathbb{Z}$. If $ab < 0$ and $a + b \geq 0$, then there exist unique integers $d, e \in \mathbb{N}$ with $d \leq e$ such that*

$$\{a, b\} = \{-d, e\}.$$

For these integers,

$$|ab| = de, \quad a + b = e - d.$$

Proof. Since $ab < 0$, exactly one of a, b is positive and the other is negative. Let e be the positive one and write the negative one as $-d$, where $d, e \in \mathbb{N}$. Then

$$a + b = e - d.$$

The assumption $a + b \geq 0$ implies $e - d \geq 0$, hence $d \leq e$. Furthermore,

$$|ab| = |-de| = de.$$

The positive member of the set $\{a, b\}$ is uniquely determined, as is the absolute value of its negative member; therefore d and e are unique. $\square$

We shall use the following direct substitution in the soundness part of the main theorem.

Lemma 3.3 (Direct verification of the two constructions). *Let $x \in \mathbb{Z}$ and let $d, e \in \mathbb{N}$.*

(a) If

$$de = x^3 - 2, \quad d + e = s^2$$

for some $s \in \mathbb{Z}_{\geq 0}$, then, for every $y \in \mathcal{Y}(s)$, both

$$(x, y, -d) \quad \text{and} \quad (x, y, -e)$$

satisfy (E).

(b) If

$$de = 2 - x^3, \quad e - d = s^2$$

for some $s \in \mathbb{Z}_{\geq 0}$, then, for every $y \in \mathcal{Y}(s)$, both

$$(x, y, d) \quad \text{and} \quad (x, y, -e)$$

satisfy (E).

Proof. In each case $y^2 = s^2$ by the definition of $\mathcal{Y}(s)$.

For part (a), if $z = -d$, then

$$\begin{aligned} z^2 + y^2z + x^3 - 2 &= d^2 - d(d + e) + de \\ &= d^2 - d^2 - de + de \\ &= 0. \end{aligned}$$

If $z = -e$, then similarly

$$\begin{aligned} z^2 + y^2z + x^3 - 2 &= e^2 - e(d + e) + de \\ &= e^2 - ed - e^2 + de \\ &= 0. \end{aligned}$$

For part (b), the identity $de = 2 - x^3$ is equivalent to $x^3 - 2 = -de$. If $z = d$, then

$$\begin{aligned} z^2 + y^2z + x^3 - 2 &= d^2 + d(e - d) - de \\ &= d^2 + de - d^2 - de \\ &= 0. \end{aligned}$$

If $z = -e$, then

$$\begin{aligned} z^2 + y^2z + x^3 - 2 &= e^2 - e(e - d) - de \\ &= e^2 - e^2 + ed - de \\ &= 0. \end{aligned}$$

This proves both assertions. □

4 The complete canonical classification

We now give the promised description of all integer solutions.

Theorem 4.1 (Complete classification). *The integer solutions of (E) are exactly the triples described in one of the following two mutually exclusive cases.*

I. The case $x \geq 2$. Choose integers d, e, s satisfying

$$d, e \in \mathbb{N}, \quad s \in \mathbb{Z}_{\geq 0}, \quad d \leq e, \quad de = x^3 - 2, \quad d + e = s^2. \quad (4)$$

Then $s > 0$, and for every $y \in \mathcal{Y}(s)$ the triples

$$(x, y, -d) \quad \text{and} \quad (x, y, -e) \quad (5)$$

are solutions. Conversely, every solution with $x \geq 2$ is of this form. For a given solution, the canonical data (d, e, s) are unique. If $d = e$, the two displayed values of z in (5) coincide and are to be retained only once.

II. The case $x \leq 1$. Choose integers d, e, s satisfying

$$d, e \in \mathbb{N}, \quad s \in \mathbb{Z}_{\geq 0}, \quad d \leq e, \quad de = 2 - x^3, \quad e - d = s^2. \quad (6)$$

Then for every $y \in \mathcal{Y}(s)$ the triples

$$(x, y, d) \quad \text{and} \quad (x, y, -e) \quad (7)$$

are solutions. Conversely, every solution with $x \leq 1$ is of this form. For a given solution, the canonical data (d, e, s) are unique. When $d = e$, one has $s = 0$, so the only value of y is 0; the two values of z are d and $-d$, which are distinct because $d > 0$.

Proof. We prove soundness, completeness, and uniqueness in each case.

Case I: $x \geq 2$. Here $x^3 - 2 \geq 6 > 0$.

Suppose first that data d, e, s satisfy (4). The asserted triples are solutions by Lemma 3.3(a). Since $d, e \in \mathbb{N}$, one has $d + e \geq 2$, and therefore $s^2 = d + e > 0$; hence $s > 0$. This proves soundness.

Conversely, let $(x, y, z) \in \mathcal{S}$ with $x \geq 2$. By Proposition 2.2, define

$$a = -z, \quad b = z + y^2.$$

Then

$$ab = x^3 - 2 > 0, \quad a + b = y^2 \geq 0.$$

Lemma 3.1 gives $a, b > 0$. Put

$$d = \min\{a, b\}, \quad e = \max\{a, b\}, \quad s = |y|.$$

Then $d, e \in \mathbb{N}$, $d \leq e$, and

$$de = ab = x^3 - 2, \quad d + e = a + b = y^2 = s^2.$$

Moreover, $z = -a$, so $z = -d$ or $z = -e$. Thus the given solution appears in (5). This proves completeness.

For uniqueness, Proposition 2.2 shows that the ordered pair $(a, b) = (-z, z + y^2)$ is uniquely determined by the solution. Hence its canonically ordered version $(d, e) = (\min\{a, b\}, \max\{a, b\})$ is unique, and $s = |y|$ is unique. If $d = e$, then $-d = -e$, so the two displayed choices of z are the same; if $d < e$, they are distinct.

Case II: $x \leq 1$. Here $x^3 - 2 \leq -1 < 0$, or equivalently $2 - x^3 > 0$.

Suppose first that data d, e, s satisfy (6). The asserted triples are solutions by Lemma 3.3(b). This proves soundness.

Conversely, let $(x, y, z) \in \mathcal{S}$ with $x \leq 1$, and again put

$$a = -z, \quad b = z + y^2.$$

Proposition 2.2 gives

$$ab = x^3 - 2 < 0, \quad a + b = y^2 \geq 0.$$

By Lemma 3.2, there exist unique $d, e \in \mathbb{N}$ with $d \leq e$ and

$$\{a, b\} = \{-d, e\}.$$

It follows that

$$de = |ab| = 2 - x^3, \quad e - d = a + b = y^2.$$

With $s = |y|$, we therefore have $e - d = s^2$. Since $z = -a$, there are precisely two possibilities:

$$a = -d \implies z = d, \quad a = e \implies z = -e.$$

Thus the given solution appears in (7), proving completeness.

Uniqueness of d, e follows from Lemma 3.2, or directly from the unique ordered pair (a, b) supplied by Proposition 2.2; again $s = |y|$ is unique. If $d = e$, then $s^2 = e - d = 0$, so $s = 0$ and $y = 0$. The corresponding values $z = d$ and $z = -e = -d$ are distinct because $d \in \mathbb{N}$.

Finally, every integer x satisfies exactly one of $x \geq 2$ and $x \leq 1$. Thus the two cases together are exhaustive and mutually exclusive. $\square$

Remark 4.2 (Logical content of the theorem). Theorem 4.1 contains all three ingredients required for a genuine classification:

- (i) every listed triple satisfies the original equation;
- (ii) every integer solution is recovered by the construction;
- (iii) the canonical factor data attached to a solution are unique, with all possible presentational coincidences explicitly identified.

No unproved existence assumption is used in any direction.

5 Equivalent compact formulations and symmetry

The canonical form of Theorem 4.1 is convenient for sign analysis. The following ordered-divisor form is more compact.

Corollary 5.1 (Ordered-divisor parametrization). *A triple $(x, y, z) \in \mathbb{Z}^3$ satisfies (E) if and only if there exist $a \in \mathbb{Z} \setminus \{0\}$ and $s \in \mathbb{Z}_{\geq 0}$ such that*

$$a \mid x^3 - 2, \quad a + \frac{x^3 - 2}{a} = s^2, \quad y \in \mathcal{Y}(s), \quad z = -a. \quad (8)$$

For a given solution, the parameter a is uniquely $a = -z$, and $s = |y|$.

Proof. Suppose (x, y, z) is a solution. By Lemma 2.1, $z \neq 0$, so $a = -z$ is nonzero. Proposition 2.2 gives

$$a(z + y^2) = x^3 - 2,$$

so $a \mid x^3 - 2$, and

$$z + y^2 = \frac{x^3 - 2}{a}.$$

As $z = -a$, this becomes

$$a + \frac{x^3 - 2}{a} = y^2.$$

Taking $s = |y|$ gives (8).

Conversely, suppose (8) holds. Put $b = (x^3 - 2)/a$. Then $ab = x^3 - 2$ and $a + b = s^2 = y^2$. The inverse map Ψ in Proposition 2.2 therefore shows that $(x, y, -a)$ is a solution. Uniqueness of $a = -z$ and $s = |y|$ is immediate. $\square$

For fixed x and y , the equation is quadratic in z . Its two roots are interchanged by a simple involution.

Proposition 5.2 (Root-swapping involution). *Define*

$$\iota : \mathcal{S} \longrightarrow \mathcal{S}, \quad \iota(x, y, z) = (x, y, -y^2 - z).$$

Then ι is an involution: ι^2 is the identity on $\mathcal{S}$. Under the bijection of Proposition 2.2, ι exchanges the ordered factors a and b .

A solution is fixed by ι if and only if

$$2z + y^2 = 0,$$

equivalently if and only if its associated factors satisfy $a = b$. Such a fixed point can occur only in the case $x \geq 2$.

Proof. Let $(x, y, z) \in \mathcal{S}$ and put $z' = -y^2 - z$. Then

$$\begin{aligned} (z')^2 + y^2z' &= (-y^2 - z)^2 + y^2(-y^2 - z) \\ &= (y^2 + z)^2 - y^2(y^2 + z) \\ &= z(y^2 + z) \\ &= z^2 + y^2z. \end{aligned}$$

Therefore

$$(z')^2 + y^2z' + x^3 - 2 = z^2 + y^2z + x^3 - 2 = 0,$$

so ι maps $\mathcal{S}$ to itself. Applying the map twice gives

$$-y^2 - (-y^2 - z) = z,$$

so ι^2 is the identity.

For the factor correspondence, if $a = -z$ and $b = z + y^2$, then for $z' = -y^2 - z$ one has

$$-z' = y^2 + z = b, \quad z' + y^2 = -z = a.$$

Thus a and b are exchanged.

The fixed-point condition $z' = z$ is exactly $-y^2 - z = z$, or $2z + y^2 = 0$. Since ι exchanges a and b , this is also equivalent to $a = b$. But $a = b$ implies $ab = a^2 > 0$ because $a \neq 0$ by Lemma 2.1. Hence $x^3 - 2 = ab > 0$, which forces $x \geq 2$. $\square$

6 Effective enumeration for fixed x

The classification immediately yields a finite, terminating procedure for each fixed integer x .

Proposition 6.1 (Complete divisor algorithm). *For a fixed $x \in \mathbb{Z}$, all elements of $\mathcal{S}_x$ are obtained by the following finite procedure.*

1. If $x \geq 2$, set $N = x^3 - 2$. Enumerate every positive divisor d of N satisfying $d \leq N/d$, and put $e = N/d$. Retain the pair (d, e) exactly when $d + e$ is a square, say $d + e = s^2$. For each retained pair output

$$(x, y, -d), \quad (x, y, -e) \quad (y \in \mathcal{Y}(s)),$$

with repeated triples removed.

2. If $x \leq 1$, set $N = 2 - x^3$. Enumerate every positive divisor d of N satisfying $d \leq N/d$, and put $e = N/d$. Retain the pair (d, e) exactly when $e - d$ is a square, say $e - d = s^2$. For each retained pair output

$$(x, y, d), \quad (x, y, -e) \quad (y \in \mathcal{Y}(s)).$$

The procedure terminates, outputs only solutions, and outputs every solution with the prescribed value of x .

Proof. In either case, N is a positive integer, so it has only finitely many positive divisors. The enumeration therefore terminates. Every output is a solution by Lemma 3.3. Conversely, the completeness part of Theorem 4.1 says that every solution in $\mathcal{S}_x$ has exactly one of the retained canonical factor pairs. Hence no solution is omitted. $\square$

For completeness, we record an exact counting formula in terms of admissible factor pairs and a simple universal bound for each fiber.

Corollary 6.2 (Finiteness and a divisor bound). *For every $x \in \mathbb{Z}$, the set $\mathcal{S}_x$ is finite. Define the set of admissible canonical factor pairs by*

$$\mathcal{Q}_x = \begin{cases} \{(d, e) \in \mathbb{N}^2 : d \leq e, de = x^3 - 2, d + e \text{ is a square}\}, & x \geq 2, \\ \{(d, e) \in \mathbb{N}^2 : d \leq e, de = 2 - x^3, e - d \text{ is a square}\}, & x \leq 1. \end{cases}$$

Then

$$\#\mathcal{S}_x = 4 \#\{(d, e) \in \mathcal{Q}_x : d < e\} + 2 \#\{(d, d) \in \mathcal{Q}_x\}. \quad (9)$$

If $\tau(n)$ denotes the number of positive divisors of n , then

$$\#\mathcal{S}_x \leq 2\tau(|x^3 - 2|). \quad (10)$$

Proof. Finiteness follows at once from Proposition 6.1.

Let $(d, e) \in \mathcal{Q}_x$. If $d < e$, then the relevant square is positive: for $x \geq 2$ it is $d + e > 0$, and for $x \leq 1$ it is $e - d > 0$. Hence there are two values of y . There are also two distinct values of z , because $-d \neq -e$ in the first case and $d \neq -e$ in the second. Thus the pair contributes exactly four solutions.

If $d = e$, then the pair contributes exactly two solutions. Indeed, for $x \geq 2$ the two displayed values of z coincide, while $d + e = 2d > 0$ gives two values of y whenever the pair

is admissible. For $x \leq 1$, one has $e - d = 0$, so $y = 0$ is the only value, while the two values of z are d and $-d$. The uniqueness assertion in Theorem 4.1 ensures that contributions from distinct canonical pairs do not overlap. This proves (9).

Put $n = |x^3 - 2|$. The positive factor pairs $d \leq e$ with $de = n$ are the orbits of the positive divisors under $d \mapsto n/d$. If n is not a square, then $\tau(n) = 2r$ and there are r pairs, all with $d < e$; even if all were admissible, (9) would give at most $4r = 2\tau(n)$ solutions. If n is a square, then $\tau(n) = 2r + 1$; there are r pairs with $d < e$ and one diagonal pair. The maximum possible contribution is then

$$4r + 2 = 2(2r + 1) = 2\tau(n).$$

This proves (10). □

7 Illustrative examples

The following table displays selected admissible factor pairs and the solutions contributed by each pair. Each row is justified immediately by Theorem 4.1; no separate search in z is needed.

x	Factor identity	Square identity	Solutions contributed
1	$2 - 1^3 = 1 \cdot 1$	$1 - 1 = 0^2$	$(1, 0, 1)$ and $(1, 0, -1)$
0	$2 - 0^3 = 1 \cdot 2$	$2 - 1 = 1^2$	$(0, \pm 1, 1)$ and $(0, \pm 1, -2)$
-2	$2 - (-2)^3 = 1 \cdot 10$	$10 - 1 = 3^2$	$(-2, \pm 3, 1)$ and $(-2, \pm 3, -10)$
-7	$2 - (-7)^3 = 5 \cdot 69$	$69 - 5 = 8^2$	$(-7, \pm 8, 5)$ and $(-7, \pm 8, -69)$
8	$8^3 - 2 = 15 \cdot 34$	$15 + 34 = 7^2$	$(8, \pm 7, -15)$ and $(8, \pm 7, -34)$
20	$20^3 - 2 = 31 \cdot 258$	$31 + 258 = 17^2$	$(20, \pm 17, -31)$ and $(20, \pm 17, -258)$

A single value of x can admit more than one canonical factor pair. For example,

$$26^3 - 2 = 17574 = 58 \cdot 303 = 87 \cdot 202,$$

and

$$58 + 303 = 361 = 19^2, \quad 87 + 202 = 289 = 17^2.$$

Consequently these two factor pairs yield the eight solutions

$$\begin{aligned} (26, \pm 19, -58), & \quad (26, \pm 19, -303), \\ (26, \pm 17, -87), & \quad (26, \pm 17, -202). \end{aligned}$$

The uniqueness clause of Theorem 4.1 shows that the two factor pairs account for disjoint sets of triples.

8 Conclusion

The substitution

$$a = -z, \quad b = z + y^2$$

turns (E) into the simultaneous relations

$$ab = x^3 - 2, \quad a + b = y^2,$$

and Proposition 2.2 proves that this transformation is a bijection, not merely a necessary condition. The sign of $x^3 - 2$ then forces exactly the two canonical alternatives in Theorem 4.1. Therefore the theorem gives an exhaustive, sound, and unique divisor-theoretic description of every integer solution. For each fixed x , the classification is effective and finite, as made explicit by Proposition 6.1.

The result should be understood precisely in this sense: it completely classifies the triples through an exact factor-pair criterion and a finite algorithm on every fiber $\mathcal{S}_x$. It does not require, and does not claim, a separate closed-form description of the set of all x for which $\mathcal{S}_x$ is nonempty.